

Rogers–Ramanujan and Distinct-Part Entropy Coefficients

Summary of `rr_entropy.py`

(a) Theory

The three partition classes and their entropy coefficients

The paper identifies three principal partition classes, each governing a physically distinct model:

- **Unrestricted partitions** $p(L)$ (Models M0–M2): all partitions of L . Entropy coefficient $\alpha_0 = \pi\sqrt{2/3} \approx 2.565$ from the Hardy–Ramanujan formula.
- **Gap-2 partitions** $q_{\text{RR}}(L)$ (Model M5): partitions whose consecutive parts (in non-increasing order) differ by at least 2 — the Rogers–Ramanujan class. Entropy coefficient $\alpha_{\text{RR}} = 2\pi/\sqrt{15} \approx 1.622$, a 37% reduction from α_0 .
- **Distinct-part partitions** $d(L)$ (Model M4): partitions with all parts distinct (strict partitions). Entropy coefficient $\alpha_d = \pi/\sqrt{3} \approx 1.814$, a 29% reduction from α_0 .

All three satisfy $\log X(L) \sim \alpha_X \sqrt{L}$ as $L \rightarrow \infty$, with the ratios

$$\frac{\alpha_{\text{RR}}}{\alpha_0} = \sqrt{\frac{2}{5}} \approx 0.6325, \quad \frac{\alpha_d}{\alpha_0} = \frac{1}{\sqrt{2}} \approx 0.7071$$

being exact algebraic identities derivable from the respective generating functions.

The Rogers–Ramanujan identity

The first Rogers–Ramanujan identity states

$$\sum_{n \geq 0} \frac{x^{n^2}}{(x; x)_n} = \prod_{\substack{n \geq 1 \\ n \equiv \pm 1 \pmod{5}}} \frac{1}{1 - x^n}$$

The product side shows that gap-2 partitions are equinumerous with partitions whose parts are congruent to $\pm 1 \pmod{5}$. The entropy coefficient α_{RR} follows from the Meinardus theorem applied to this product: only $2/5$ of all positive integers appear as admissible parts, which is what reduces α relative to the unrestricted case.

Convergence rate

The convergence $\log X(L)/\sqrt{L} \rightarrow \alpha_X$ is slow: the sub-leading correction is of order $\log L/\sqrt{L}$, not $1/\sqrt{L}$. At $L = 10000$ the effective coefficients are still approximately 5% below their asymptotic values. This slow approach to the limit is directly visible in the paper's convergence figures (Figs. 3–4).

(b) What the script does

The script verifies the three entropy coefficients numerically by exact enumeration, for L up to 10000. It produces:

1. **Three convergence tables:** $\log X(L)/\sqrt{L}$ versus L for $X \in \{p, q_{\text{RR}}, d\}$, showing approach to the theoretical asymptote from below, the percentage error at each L , and the ratio to α_0 .

2. **A combined ratio table:** $\alpha_X(L) \equiv \log X(L)/\sqrt{L}$ for all three classes side by side, alongside the finite- L ratios $\alpha_{RR}(L)/\alpha_p(L)$ and $\alpha_d(L)/\alpha_p(L)$, converging to $\sqrt{2/5}$ and $1/\sqrt{2}$ respectively.
3. **Three CSV files:** `rr_entropy_convergence.csv`, `distinct_part_convergence.csv`, and `entropy_coefficients.csv` (combined), used to produce the convergence figures in the paper.
4. **A summary block** at $L = 2000$: observed effective coefficients, percentage errors against theory, and observed entropy reductions.

(c) How the script does it

Unrestricted partitions: `compute_p`

Uses Euler's pentagonal number theorem recurrence:

$$p(L) = \sum_{k \geq 1} (-1)^{k+1} \left[p\left(L - \frac{k(3k-1)}{2}\right) + p\left(L - \frac{k(3k+1)}{2}\right) \right]$$

with $p(0) = 1$ and $p(n) = 0$ for $n < 0$. This is a one-dimensional recurrence running in $O(L\sqrt{L})$ time and $O(L)$ space, substantially more efficient than the two-dimensional $p_k(L)$ table used in `be_saddle.py`.

Gap-2 partitions: `compute_q_rr`

Uses a two-dimensional DP. Let $f(n, m)$ = number of gap-2 partitions of n whose largest part is $\leq m$. The recurrence is:

$$f(n, m) = f(n, m-1) + \begin{cases} f(n-m, m-2) & m \geq 2 \\ \mathbf{1}[n=1] & m = 1 \end{cases}$$

with $f(0, m) = 1$ for all $m \geq 0$. The first term counts partitions not using m as a part; the second counts those that use m exactly once (which requires the next largest part to be $\leq m-2$ to satisfy the gap-2 condition). The $m = 1$ special case arises because two parts equal to 1 would have gap 0, so $f(n, 1) = 1$ only if $n = 1$. The answer is $q_{RR}(L) = f(L, L)$, computed for all L up to $L_{\max}$ in a table of size $(L_{\max} + 1) \times (L_{\max} + 3)$ and $O(L^2)$ time.

Distinct-part partitions: `compute_d`

Uses a standard knapsack DP in which each part value $1, 2, \dots, L$ is processed once, and the array is traversed in reverse to ensure each part value is used at most once:

```
dp[0] = 1
for part in 1..L_max:
    for j in L_max..part (reverse):
        dp[j] += dp[j - part]
```

This runs in $O(L^2)$ time and $O(L)$ space.

Convergence analysis

`convergence_table` computes $\log X(L)/\sqrt{L}$ at the L values

`{5, 10, 20, 30, 50, 75, 100, 150, 200, 300, 400, 500, 750, 1000, 1250, 1500, 1750, 2000, 3000, 5000, 7500, 10000}`

and reports the percentage error against the theoretical asymptote and the ratio to α_0 . The three tables share the same L grid for direct comparison.

Spot-checks

Before the main analysis, the script verifies the DP counts against known values: $p(10) = 42$, $p(20) = 627$, $p(50) = 204,226$; $d(10) = 10$, $d(20) = 64$, $d(50) = 3,658$; and prints q_{RR} at $L = 10, 20, 50$ for manual inspection.

(d) Output produced

Convergence tables (selected rows)

L	$\log p(L)/\sqrt{L}$	$\log q_{RR}(L)/\sqrt{L}$	$\log d(L)/\sqrt{L}$
100	2.2175	1.3929	1.5572
500	2.3699	1.5026	1.6787
1000	2.4175	1.5366	1.7165
2000	2.4647	1.5699	1.7540
10000	2.5293	1.6060	1.7946
∞ (theory)	2.5651	1.6223	1.8138

All three series approach their limits from below, with the gap narrowing as $\sim \log L/\sqrt{L}$.

Combined ratio table (selected rows)

L	α_{RR}/α_p	α_d/α_p
100	0.6281	0.7020
500	0.6337	0.7083
2000	0.6369	0.7116
10000	0.6395	0.7107
∞ (theory)	0.6325	0.7071

Both ratios converge to their theoretical values from above (the effective ratio overshoots slightly at large L because α_p converges more slowly than α_{RR} and α_d).

Summary block at $L = 2000$

At L=2000:

```
alpha_p (unrestricted) = 2.46466 (theory 2.56510, err -3.92%)
alpha_RR (gap-2)       = 1.56990 (theory 1.62231, err -3.23%)
alpha_d (distinct)     = 1.75401 (theory 1.81380, err -3.30%)
```

```
Observed ratio alpha_RR/alpha_p = 0.63697 (theory sqrt(2/5) = 0.63246)
```

```
Observed ratio alpha_d/alpha_p = 0.71167 (theory 1/sqrt(2) = 0.70711)
```

Entropy reduction (RR):	3.7%	(theory 36.8%)	[finite- L value]
Entropy reduction (distinct):	28.8%	(theory 29.3%)	

(Note: the "entropy reduction" row compares $1 - \alpha_X(L)/\alpha_p(L)$ at $L = 2000$, which underestimates the asymptotic reduction because both coefficients are still below their limits.)

CSV files

Three files are written to the same directory as the script:

- `rr_entropy_convergence.csv`: columns L , $\log q_{RR}$, α_{eff} , α_{RR_theory} , pct_error , $ratio_over_alpha0$.
- `distinct_part_convergence.csv`: same structure for $d(L)$.
- `entropy_coefficients.csv`: all three models combined, with columns for $\log p$, $\log q_{RR}$, $\log d$, the three effective α values, and the two ratios.

(e) What we learn

The 37% and 29% entropy reductions are confirmed numerically. At $L = 2000$ the observed reductions are 3.9% below the asymptotic predictions, consistent with the $O(\log L/\sqrt{L})$ convergence rate. The data confirm that $\alpha_{RR}/\alpha_0 \rightarrow \sqrt{2/5}$ and $\alpha_d/\alpha_0 \rightarrow 1/\sqrt{2}$ as $L \rightarrow \infty$.

Convergence is slow and must be accounted for when comparing with simulation. At physically accessible chain lengths ($L \lesssim 500$) the effective entropy coefficients are 3–5% below their asymptotic values. The paper's convergence figures (Figs. 3–4) make this explicit, showing the data approaching the asymptotes from below across the full computed range. Any comparison with primitive-path analysis data at finite L should use the exact finite- L values from the DP table, not the asymptotic $\alpha_X\sqrt{L}$ formula directly.

The three algorithms have different computational profiles. `compute_p` is $O(L\sqrt{L})$ in time and $O(L)$ in space — fast even at $L = 10000$. `compute_q_rr` and `compute_d` both require $O(L^2)$ time; `compute_q_rr` also requires $O(L^2)$ space (the full 2D table), while `compute_d` uses $O(L)$ space. At $L_{\max} = 10000$ the `compute_q_rr` table occupies approximately 400 MB, which is the dominant memory cost of the script.

The script provides the data underlying three paper figures. The CSV files feed directly into the scripts that produce Figs. 3 (`fig_entropy_convergence.pdf`, left and right panels) and Fig. 4 (`fig_entropy_ratios.pdf`) of the paper. The spot-checks on p , d , and q_{RR} at small L confirm the DP implementations are correct before running to $L = 10000$.